

Eduardo Valentin Perez Hernandez

M.Sc. Researcher — Biosignal Analysis, Machine Learning & Pattern Recognition

M +52 223 130 89 12
E eduardo.perezherna@ieee.org
in [valent-in00](#)

Research Interests

Biosignal analysis (EEG, ECG), machine learning and pattern recognition for time-series classification, feature extraction and dimensionality reduction, representation learning, deep learning for sequential data, and natural language processing applied to neural signals.

Education

- 2024 – 2026 **M.Sc. in Computer Science**, *Instituto Nacional de Astrofísica, Óptica y Electrónica (INAOE)*, Puebla, Mexico
Thesis: “*Open-Vocabulary Text Decoding from Electroencephalographic Signals During Reading Tasks.*”
Designing and evaluating feature extraction pipelines (HHT, PSD, temporal statistics) and deep learning architectures for sequence-to-sequence decoding of biosignals; studying the effect of spatial channel configurations on classification performance.
Advisors: Dr. Alejandro A. Torres-García & Dr. Luis Villaseñor-Pineda.
- 2018 – 2024 **B.Eng. in Mechatronics Engineering**, *Benemérita Universidad Autónoma de Puebla (BUAP)*, Puebla, Mexico
Thesis: “*Shared Control System for a Holonomic Mobile Robot Based on a Brain-Computer Interface.*”
Applied convolutional neural networks (EEGNet) for real-time biosignal classification; designed a fuzzy logic shared-control system; integrated embedded hardware (Jetson Nano, Raspberry Pi Pico) with sensors and actuators.
Advisor: Dr. María Monserrat Morín Castillo.

Publications

- 2026 **A Comparative Study of EEG Representations and Channel Settings for EEG-to-Text Decoding during Reading Tasks**, *10th Graz Brain-Computer Interface Conference*, Graz, Austria, **Accepted — Travel Grant**
E.V. Pérez Hernández, A.A. Torres-García, L. Villaseñor-Pineda.
Benchmarked signal representations (HHT, PSD, temporal statistics) and spatial channel subsets for sequence-to-sequence decoding of biosignals; best result: BLEU-1 = 11.45, ROUGE-1 F1 = 11.52 (ZuCo v1&v2).
- 2025 **Intersubject Variability in Classification Models for Brain-Computer Interfaces**, *Research in Computing Science*, Vol. 154(10), pp. 143–154
E.V. Pérez Hernández, M. Aviña-Corral, Z.A. Gárate-Cahuantzi, J.A. Carrasco-Ochoa, J.F. Martínez-Trinidad, A.A. Torres-García.
Studied the effect of subject variability on pattern recognition models; compared subject-specific, multisubject, and cross-subject training schemes using CSP + LDA/SVM; subject-specific model reached 90.71%.

- Jun 2025 **Evaluation of Spatial and Frequency Features for Improving the Classification of Motor Imagery**, *11th International Brain-Computer Interface Meeting*, Brussels, Belgium, **Poster**
E.V. Pérez Hernández, M. Aviña-Corral, A.A. Torres-García, J.A. Carrasco-Ochoa, J.F. Martínez-Trinidad.
Combined spatial (CSP) and frequency (DWT) feature extraction for biosignal classification; applied Relief-F for dimensionality reduction; EEGNet+DWT achieved 71.13% average accuracy.
DOI: 10.3217/978-3-99161-050-2
- Sep 2025 **Towards a Neurotutor: Comparative Analysis of Machine Learning Models for EEG-Based Reading Task Classification**, *IEEE Brain Discovery & Neurotechnology Workshop*, Vancouver, BC, Canada, **Poster — Travel Grant**
Applied and compared supervised learning classifiers for time-series biosignal classification; evaluated generalization across cognitive tasks and subjects.
- 2024 **Offline Study of a BCI for Holonomic Mobile Robot Control**, *ACONTACS, Vol. 6, pp. 18–25*
E.V. Pérez Hernández, M.M. Morin Castillo, J.R. Conde Sánchez.
Applied deep learning (EEGNet) for four-class biosignal classification; validated the pattern recognition pipeline offline before embedded deployment. ISSN: 2992-6750.

Awards & Grants

- Sep 2026 **Student Travel Grant**, *10th Graz Brain-Computer Interface Conference*, Graz, Austria
Competitive travel grant awarded to present research at the premier international BCI conference.
- Sep 2025 **Student Travel Grant**, *IEEE Brain Discovery & Neurotechnology Workshop*, Vancouver, BC, Canada
Competitive travel grant awarded to attend and present at the IEEE Brain Discovery workshop.
- 2024 **Admission by Outstanding Academic Performance**, *INAOE — M.Sc. Computer Science Program*, Puebla, Mexico
Admitted to the master's program on the basis of distinguished undergraduate academic performance.

Research Experience

- Oct 2025 – Oct 2026 **Research Fellow**, *Laboratorio Nacional de Supercómputo del Sureste de México (LNS)*, Puebla, Mexico
Project: “Open-Vocabulary Text Decoding from Electroencephalographic Signals During Natural Reading.” (No. 202504065C)
Developing and training deep learning models for biosignal decoding on HPC infrastructure; leveraging GPU clusters and supercomputing resources for large-scale experimentation.
- Jun – Nov 2022 **Research Assistant**, *Benemérita Universidad Autónoma de Puebla*, Puebla, Mexico
Contributed to theoretical analysis for on-chip signal processing and classification of electrocardiographic (ECG) biosignals.
CONACYT-funded project (No. 319601).

Skills

Programming	Python, MATLAB, Simulink, C++
ML / DSP	PyTorch, Scikit-learn, MNE-Python, HuggingFace Transformers, SciPy
Hardware	FPGA, Microcontrollers (Arduino, Raspberry Pi, Jetson Nano), Embedded Systems, IoT
HPC / Tools	GPU computing, Supercomputing (LNS), Git, ROS, $\LaTeX$
Languages	Spanish (Native), English (B1)

Leadership & Service

- 2023 – 2025 **President, Student Activities Committee, IEEE Puebla Section**
Coordinated events, initiatives, and student engagement across the regional IEEE student community.
- 2025 **Secretary, Computer Society Professional Chapter, IEEE Puebla Section**
Supported chapter governance, communications, and professional development activities.
- 2025 **Webmaster & Social Media Chair, IEEE Mexican Humanitarian Technology Conference (MHTC) 2025**
Managed the conference website and social media outreach.
- 2022 **President, Student Branch, IEEE — Benemérita Universidad Autónoma de Puebla**
Organized technical workshops, talks, and community-building activities for engineering students.